

命题逻辑的推理 I

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学习内容

命题逻辑的推理：

A. 几种证明方法

B. 模型检验

C. 逻辑等价

D. 基于推理规则

E. 归结推理

F. Horn子句和限定子句

G. 前向链接和反向链接



A 几种证明方法

证明方法

1. $KB \models \alpha$ iff $M(KB) \subseteq M(\alpha)$
2. $KB \models \alpha$ iff $(KB \wedge \neg \alpha)$ is unsatisfiable
3. $KB \models \alpha$ iff $KB \Rightarrow \alpha$ is valid
4. $KB \models \alpha$ iff $KB, \neg \alpha \models \text{False}$

基于模型
基于可满足性
基于有效性
基于反演推理

□ to prove $KB \models \alpha$:

- for any m in $M(KB)$, check whether it is also in $M(\alpha)$. (1)
- for any m , $KB \wedge \neg \alpha$ is false. (2)
- for any m , $KB \Rightarrow \alpha$ is true. (3)
- $KB \wedge \neg \alpha$ is logically equivalent to False. (2)
- $KB \Rightarrow \alpha$ is logically equivalent to True. (3)
- From KB , derive α . (from definition)
- From $KB, \neg \alpha$, derive False. (4)

Model checking
truth table enumeration

logical
equivalence

inference rules
resolution



B 模型检验(MODEL CHECKING)

模型检验

□ 模型检验：

- 枚举所有可能的模型，检验 KB 为真的情况下 α 是否都为真，即判断 $M(KB) \subseteq M(\alpha)$ 是否成立
- 枚举所有可能的模型，检验 $KB \Rightarrow \alpha$ 是否为真
- 枚举所有可能的模型，检验 $KB \wedge \neg\alpha$ 是否为假

模型检验

• detecting nothing in [1,1], moving right, breeze in [2,1]

☐ Pits cause breezes in adjacent squares

■ $R_1: B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$

■ $R_2: B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$

☐ Start

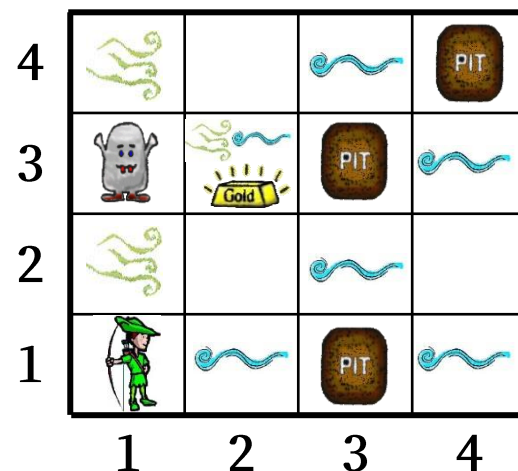
■ $R_3: \neg P_{1,1}$

☐ Perceiving

■ $R_4: \neg B_{1,1}$

■ $R_5: B_{2,1}$

☐ is $\neg P_{1,2}$ true????



☐ $KB \models \neg P_{1,2} ?$



请找出让KB为真的所有模型

$B_{1,1}$	$B_{2,1}$	$P_{1,1}$	$P_{1,2}$	$P_{2,1}$	$P_{2,2}$	$P_{3,1}$
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模型检验

•detecting nothing in [1,1], moving right, breeze in [2,1]

□ Pits cause breezes in adjacent squares

■ $R_1: B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$

■ $R_2: B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$

□ Start

■ $R_3: \neg P_{1,1}$

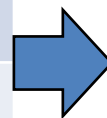
□ Perceiving

■ $R_4: \neg B_{1,1}$

■ $R_5: B_{2,1}$

□ is $\neg P_{1,2}$ true?

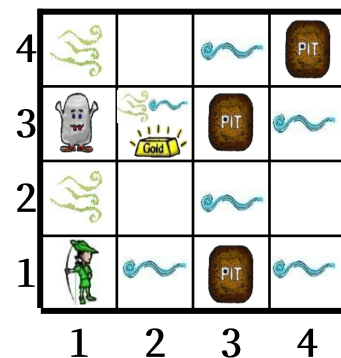
$B_{1,1}$	$B_{2,1}$	$P_{1,1}$	$P_{1,2}$	$P_{2,1}$	$P_{2,2}$	$P_{3,1}$
F	T	F	F	F	F	T
F	T	F	F	F	T	F
F	T	F	F	F	T	T



□ $KB \models \neg P_{1,2}$

模型检验

□ $KB \models \neg P_{1,2}$?



$B_{1,1}$	$B_{2,1}$	$P_{1,1}$	$P_{1,2}$	$P_{2,1}$	$P_{2,2}$	$P_{3,1}$	R_1	R_2	R_3	R_4	R_5	KB
false	false	false	false	false	false	false	true	true	true	true	false	false
false	false	false	false	false	false	true	true	true	false	true	false	false
...	...	...	...	...	...	...	...	...	...	...	...	...
false	true	false	false	false	false	false	true	true	false	true	true	false
false	true	false	false	false	false	true	true	true	true	true	true	true
false	true	false	false	false	true	false	true	true	true	true	true	true
false	true	false	false	false	true	true	true	true	true	true	true	true
false	true	false	false	true	false	false	true	false	false	true	true	false
...	...	...	...	...	...	...	...	...	...	...	...	...
true	true	true	true	true	true	true	false	true	true	false	true	false

模型检验

□ $(A \vee \neg B) \wedge (B \vee C) \Rightarrow A \vee C$ valid or not?

A	B	C	$A \vee \neg B$	$B \vee C$	$A \vee C$	$(A \vee \neg B) \wedge (B \vee C) \Rightarrow A \vee C$
false	false	false	true	false	false	true
false	false	true	true	true	true	true
false	true	false	false	true	false	true
false	true	true	false	true	true	true
true	false	false	true	false	true	true
true	false	true	true	true	true	true
true	true	false	true	true	true	true
true	true	true	true	true	true	true

蕴涵的真值表枚举算法

function TT-ENTAILS?(KB, α) **returns** true or false

inputs: KB, the knowledge base, a sentence in propositional logic

α , the query, a sentence in propositional logic

symbols $\leftarrow$ a list of the proposition symbols in KB and α

return TT-CHECK-ALL(KB, α , symbols,{ })

function TT-CHECK-ALL(KB, α , symbols ,model) **returns** true or false

//symbols: 还没有赋值的符号; model: 目前的部分赋值

if EMPTY?(symbols) **then**

//所有符号都已赋值

if PL-TRUE?(KB,model) **then return** PL-TRUE?(α ,model)

else return true // when KB is false, always return true

else do

P $\leftarrow$ FIRST(symbols)

rest $\leftarrow$ REST(symbols)

return (TT-CHECK-ALL(KB, α , rest ,model $\cup$ {P = true})

and

TT-CHECK-ALL(KB, α , rest ,model $\cup$ {P = false })))

枚举所有可能的模型，检验KB为真时 α 是否为真。
(枚举所有可能的模型，检验 $KB \Rightarrow \alpha$ 是否一直为真)

推理算法的可靠性和完备性

- $KB \vdash_i \alpha$: 推理算法i可以从KB推理出语句 α .
- *Soundness*(可靠性): 对于推理算法i从KB推理出的每条语句 α , 都有 $KB \models \alpha$, 则i是可靠的
- *Completeness*(完备性): 对于KB蕴涵的每条语句 α , 推理算法i都能从KB推理出 α , 则i是完备的
- TT-Entails?(KB, α)是可靠的、完备的
真值表枚举算法



C 逻辑等价

逻辑等价

□ Logical equivalence:

- $\alpha \equiv \beta$ iff they are true in the same set of models
- $\alpha \equiv \beta$ iff $\alpha \models \beta$ and $\beta \models \alpha$.

1. $(\alpha \wedge \beta) \equiv (\beta \wedge \alpha)$ commutativity of $\wedge$

2. $(\alpha \vee \beta) \equiv (\beta \vee \alpha)$ commutativity of $\vee$

3. $((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma))$ associativity of $\wedge$

4. $((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma))$ associativity of $\vee$

5. $\neg(\neg\alpha) \equiv \alpha$ double-negation elimination

6. $(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha)$ contraposition

7. $(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta)$ implication elimination

8. $(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha))$ biconditional elimination

9. $\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta)$ de Morgan

10. $\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta)$ de Morgan

11. $(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma))$ distributivity of $\wedge$ over $\vee$

12. $(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma))$ distributivity of $\vee$ over $\wedge$

通过逻辑等价证明蕴涵

1. $KB \models \alpha$ iff $KB \wedge \neg\alpha$ is unsatisfiable
 - 证明 $KB \wedge \neg\alpha \equiv \text{False}$
2. $KB \models \alpha$ iff $KB \Rightarrow \alpha$ is valid
 - 证明 $KB \Rightarrow \alpha \equiv \text{True}$

通过逻辑等价证明蕴涵

□ 证明 $(A \vee \neg B) \wedge (B \vee \neg C) \models A \vee \neg C$

□ 证明 $(A \vee \neg B) \wedge (B \vee \neg C) \Rightarrow A \vee \neg C$ 是有效的

$$\begin{aligned}
 & (A \vee \neg B) \wedge (B \vee \neg C) \Rightarrow A \vee \neg C \\
 \equiv & \neg((A \vee \neg B) \wedge (B \vee \neg C)) \vee A \vee \neg C \\
 \equiv & (\neg(A \vee \neg B) \vee \neg(B \vee \neg C)) \vee A \vee \neg C \\
 \equiv & (\neg A \wedge B) \vee (\neg B \wedge C) \vee A \vee \neg C \\
 \equiv & (\neg A \wedge B) \vee A \vee (\neg B \wedge C) \vee \neg C \\
 \equiv & ((\neg A \vee A) \wedge (B \vee A)) \vee ((\neg B \vee \neg C) \wedge (C \vee \neg C)) \\
 \equiv & (B \vee A) \vee (\neg B \vee \neg C) \\
 \equiv & \text{True} \vee A \vee \neg C \\
 \equiv & \text{True}
 \end{aligned}$$

$$\begin{aligned}
 \alpha \wedge \text{True} &\equiv \alpha \\
 \alpha \wedge \text{False} &\equiv \text{False} \\
 \alpha \vee \text{True} &\equiv \text{True} \\
 \alpha \vee \text{False} &\equiv \alpha
 \end{aligned}$$

1. $(\alpha \wedge \beta) \equiv (\beta \wedge \alpha)$ commutativity of $\wedge$
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3. $((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma))$ associativity of $\wedge$
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5. $\neg(\neg\alpha) \equiv \alpha$ double-negation elimination
6. $(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha)$ contraposition
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8. $(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha))$ biconditional elimination
9. $\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta)$ de Morgan
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D 基于推理规则

推理规则

□ Application of inference rules

- Legitimate generation of new sentences from old
old sentences
new sentences
- **Proof = a sequence of inference rule applications**
- Can use inference rules as operators in a standard search algorithm
- Typically require transformation of sentences into a **normal form**

推理规则

□ Modus Ponens (假言推理规则) $\frac{\alpha \Rightarrow \beta, \alpha}{\beta}$

□ And-Elimination (消去合取词) $\frac{\alpha \wedge \beta}{\alpha}$

□ logical equivalence

$$\alpha \Leftrightarrow \beta \equiv (\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)$$

$$\frac{\alpha \Leftrightarrow \beta}{(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)} \quad \frac{(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)}{\alpha \Leftrightarrow \beta}$$

□ 假言三段论: $\frac{\alpha \Rightarrow \beta, \beta \Rightarrow \gamma}{\alpha \Rightarrow \gamma}$

□ 析取引入: $\frac{\alpha}{\alpha \vee \beta}$



使用推理规则的实例

推理规则

□ Pits cause breezes in adjacent squares

$$\blacksquare R_1: B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$$

$$\blacksquare R_2: B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$$

□ Start

$$\blacksquare R_3: \neg P_{1,1}$$

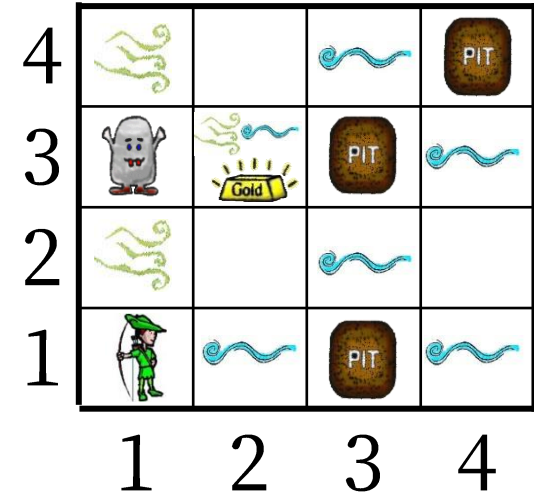
□ Perceiving

$$\blacksquare R_4: \neg B_{1,1}$$

$$\blacksquare R_5: B_{2,1}$$

□ $KB \models \neg P_{1,2} ?$

□ From KB, how to derive $\neg P_{1,2}$ based on the rules:



Modus Ponens

$$\frac{\alpha \Rightarrow \beta, \alpha}{\beta}$$

And-Elimination

$$\frac{\alpha \wedge \beta}{\alpha}$$

1. $(\alpha \wedge \beta) \equiv (\beta \wedge \alpha)$ commutativity of $\wedge$

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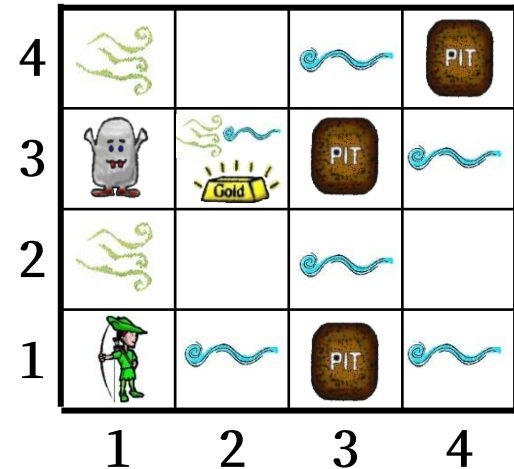
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12. $(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma))$ distributivity of $\vee$ over $\wedge$

推理规则

- Pits cause breezes in adjacent squares
 - $R_1: B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$
 - $R_2: B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$
- Start
 - $R_3: \neg P_{1,1}$
- Perceiving
 - $R_4: \neg B_{1,1}$
 - $R_5: B_{2,1}$
- $KB \models \neg P_{1,2} ?$



- $R_6: (B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$
 R_2 , biconditional-elimination
- $R_7: ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$ R_6 , And-Elimination
- $R_8: (\neg B_{1,1} \Rightarrow \neg (P_{1,2} \vee P_{2,1}))$ R_7 , contraposition
- $R_9: \neg (P_{1,2} \vee P_{2,1})$ R_8, R_4 , Modus-ponens
- $R_{10}: \neg P_{1,2} \wedge \neg P_{2,1}$ R_9 , De Morgan
- $R_{11}: \neg P_{1,2}$ R_{10} , And-Elimination



基于推理规则的方法的可靠性和完备性？

可靠性和完备性

- 可靠性：取决于推理规则。
 - 若每条推理规则都是可靠的，则推理算法是可靠的
 - 如何判断一条推理规则 $\frac{\text{old sentences}}{\text{new sentences}}$ 是否是可靠的？
- 完备性：取决于推理规则集合。



从手工证明到自动证明

Application of inference rules

- Legitimate generation of new sentences from old
old sentences
new sentence
- **Proof = a sequence of inference rule applications**
- Can use inference rules as operators in a standard search algorithm
- Typically require transformation of sentences into a **normal form**

通过搜索找到证明

- KB: Set of sentences
- I: Set of inference rules
- **Inference** is the process of applying successive inference rules from I to KB, each rule adding its conclusion to KB
- The previous is **by hand!**
- How to apply **a search algorithm?**
 - what are initial state, actions, result, goal?

形式化为一个搜索问题

- apply a search algorithm
 - Initial state: the initial KB
 - Actions: inference rules
 - Result: add inferred sentences to KB
 - Goal: the sentence we are trying to prove is in the KB.

Thanks